

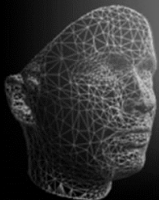
Advanced Topics in Computer Graphics II

Geometry Processing

Discrete Exterior Calculus



December 3, 2024



Example: Differential equation

Given: 2-form $\omega := dx_1 \wedge dx_2$ on $\mathbb{R}^2$

Find: 1-form α such that $d\alpha = \omega$.

Let $\alpha = udx_1 + vdx_2$ for some pair of coordinate functions $u, v : \mathbb{R}^2 \rightarrow \mathbb{R}$.

We want to find u, v such that

$$d\alpha = du \wedge dx_1 + dv \wedge dx_2 = dx_1 \wedge dx_2$$

Recalling that $dx_1 \wedge dx_2 = -dx_2 \wedge dx_1$, we must have $v = \frac{1}{2}x_1$ and $u = -\frac{1}{2}x_2$.

Therefore, we get

$$\alpha = \frac{1}{2}(x_1 dx_2 - x_2 dx_1).$$



- ▶ Solving even very easy differential equations by hand can be hard!
- ▶ If equations involve data, forget about solving them by hand!
- ▶ Instead, we need a way to approximate solutions via computation
- ▶ **Basic idea:**
 - ▶ replace domain with mesh
 - ▶ replace differential forms with values on mesh
 - ▶ replace differential operators with matrices



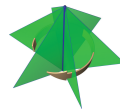
- ▶ New and more general terminology, which can describe problems in higher dimensions as well.
 - ▶ k -Simplices instead of triangles, edges and vertices
 - ▶ Simplicial complexes instead of meshes.
 - ▶ (Co)chains for defining linear combinations of simplices/differential forms.
 - ▶ The definitions of most operators work for k -simplices/ k -forms and do not need to be defined for each dimension separately
- ▶ Points of interest are located at the mesh elements. 3D physics example: Circulation preserving fluids¹.
 - ▶ The amount of fluid is stored in a 3D cell
 - ▶ The fluid flows through boundary faces of a cell into other cells
 - ▶ The divergence is the amount of fluid flowing in or out of the cell
 - ▶ The curl is the circulation of the fluid around an edge



(a) Flux is defined at the faces



(b) Divergence is defined in the cells

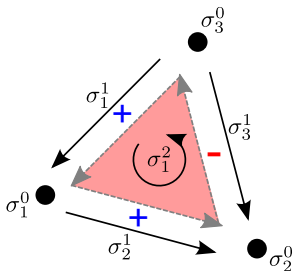


(c) Curl is defined at the edges

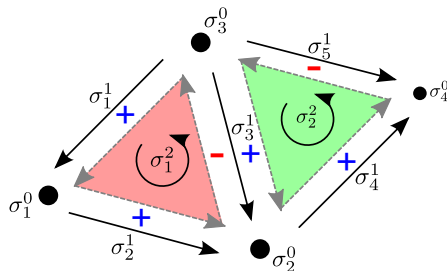
¹S. Elcott et al. "Stable, circulation-preserving, simplicial fluids". In: *ACM Transactions on Graphics (TOG)* 26.1 (2007), 4–es.



- ▶ We have seen simplices in an earlier lecture.
- ▶ In DEC notation we treat simplices as ordered $(k + 1)$ -tuples σ^k of vertices v_i .
- ▶ All elements are represented explicitly, i.e., we use for example an explicit list of edges instead of relying on the half-edges of the triangles.
- ▶ Each simplex is stored with an (arbitrary) orientation and DEC operators contain the relative orientation between the stored elements (cf. adjacency matrices).



(a) A oriented 2-simplex with its the adjacent 1-simplices and 0-simplices



(b) The 1-simplex σ_3^1 has a different relative orientation to the 2-simplices σ_1^2 and σ_2^2



Recall the definition of a simplicial complex:

- ▶ A simplicial complex is a set of simplices, for which holds
 - ▶ If the complex contains a k -simplex σ^k with $k > 0$, it also contains all its ℓ -faces for with $\ell < k$, i.e., the boundary of a simplex must also be in the simplicial complex.
 - ▶ Two k -simplices either do not intersect or intersect at a common face.
 - ▶ Neighboring simplices σ^k and σ^ℓ have a common face $\sigma^k \cap \sigma^\ell \neq \emptyset$.



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- ▶ In 2D, the definition describes a triangle mesh and in 3D a tetrahedral mesh
- ▶ The k -skeleton K^k of a simplicial complex K is the set of all k -simplices in K , e.g., $K^2 = \{\sigma_1^2, \dots, \sigma_{N_2}^2\}$ is the set of all triangles.
- ▶ The k -skeletons are stored explicitly, so each k -simplex has a unique index that allows us to look it up in the list of k -simplices K^k .
- ▶ With explicit indices and fixed orientation for all elements, signed adjacency matrices can be used for neighborhood queries.
- ▶ We will see, that these matrices also define important operators.



- ▶ We can describe a weighted subcomplex by a k -chain $c^k \in C^k$, which is a function $c^k : K^k \rightarrow \mathbb{R}$, that assigns a real value to each k -simplex σ^k .
- ▶ Because we have a finite number of simplices, a chain can be described by a vector $\mathbf{c}^k = (c^k(\sigma_1^k), \dots, c^k(\sigma_{N_k}^k)) \in \mathbb{R}^{N_k}$, which stores the values for all k -simplices.
- ▶ Example: A submesh can be specified by a chain, which has a 1 for a triangle that is part of the submesh and a 0 otherwise

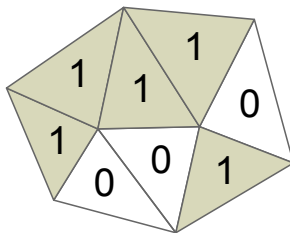


Figure: The 2-chain $c^2 = (1, 1, 1, 0, 1, 0, 0, 1)$ describes a submesh

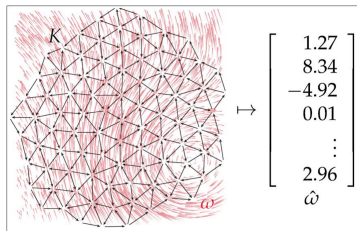


Discretization of forms: de Rham Map

Let K be an oriented simplicial complex on $\mathbb{R}^n$, and let ω be a differential k -form on $\mathbb{R}^n$. For each simplex $\sigma \in K$, the corresponding value of the discrete k -form $\hat{\omega}$ is given by

$$\hat{\omega}_\sigma := \int_\sigma \omega$$

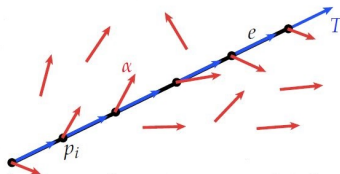
The map from continuous forms to discrete forms is called a **discretization map**, or sometimes the **de Rham map**.



Key idea: discretization just means: integrate a k -form over k -simplices. The result is just a list of values.

Basic recipe:

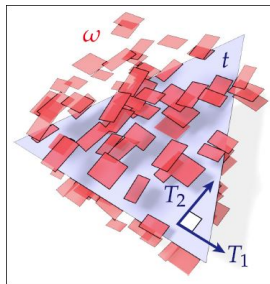
- Compute unit tangent T
- Apply α to T , yielding function $\alpha(T)$
- Integrate this scalar function over the edge



$$\begin{aligned}\hat{\alpha}_e &:= \int_e \alpha = \int_0^L \alpha(T) ds \\ &\approx \text{length}(e) \left(\frac{1}{N} \sum_{i=1}^N \alpha_{p_i}(T) \right)\end{aligned}$$

Basic recipe:

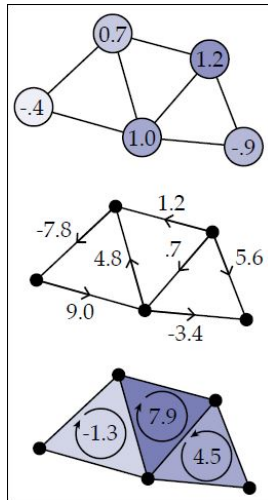
- Compute orthonormal basis T_1, T_2 for triangle t
- Apply ω to T_1, T_2 , yielding function $\omega(T_1, T_2)$
- Integrate this scalar function over the triangle



$$\begin{aligned}\hat{\omega} &:= \int_t \omega = \int_t \omega(T) ds \\ &\approx \text{area}(t) \left(\frac{1}{N} \sum_{i=1}^N \omega_{p_i}(T_1, T_2) \right)\end{aligned}$$



- ▶ Abstractly, a discrete differential k -form is just any assignment of a real value to each oriented k -simplex.
- ▶ In $2D$:
 - ▶ values at vertices encode a discrete 0-form
 - ▶ values at edges encode a discrete 1-form
 - ▶ values at faces encode a discrete 2-form
- ▶ Conceptually, values represent integrated k -forms
- ▶ In practice, almost never comes from direct integration!
- ▶ More typically, values start at vertices (samples of some function); 1-forms, 2-forms, etc., arise from applying operators like the (discrete) exterior derivative



Integration

- ▶ A (smooth) differential form in a simplicial complex is a function $\hat{\omega}^k \in \Omega^k$ that can be integrated over simplices σ^k and using a weighted sum also over k -chains as $\sum_i c^k(\sigma_i^k) \int_{\sigma_i^k} \hat{\omega}^k$.
- ▶ The smooth differential form can be discretized as a co-chain, which is a vector $\omega^k \in \mathbb{R}^k = (\int_{\sigma_0^k} \hat{\omega}^k, \dots, \int_{\sigma_{N_k}^k} \hat{\omega}^k)$.
- ▶ Examples
 - ▶ A differential form ω^0 is a function on vertices and the corresponding co-chain is the evaluation of the function on all vertices of the mesh: $\omega^0 = (\hat{\omega}^0(\sigma_0^0), \dots, \hat{\omega}^0(\sigma_{N_0}^0))$
 - ▶ A differential form $\hat{\omega}^1$ can be integrated over 1-simplices and the corresponding discrete form is the evaluation the mesh edges: $\omega^1 = (\int_{\sigma_0^1} \hat{\omega}^1, \dots, \int_{\sigma_{N_1}^1} \hat{\omega}^1)$

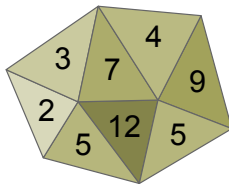


Figure: The 2-co-chain $\omega^2 = (3, 7, 4, 9, 2, 5, 12, 5)$ assigns a value to each triangle



- ▶ The evaluation of the co-chain ω^k on a chain c^k can now be defined as the scalar product $\omega^k c^k := \langle \omega^k, c^k \rangle$
- ▶ The integration is discrete, i.e., a integration domain cannot contain only a part of a simplex
 - ▶ But simplices can be weighted by by using values other than 1, so we can integrate over “half a simplex”, but we cannot specify the part over which is integrated ...
 - ▶ ... but the value does not vary on a simplex anyway.
- ▶ the volume form $\mathbb{1}$ can be defined as:
 - ▶ $\langle \mathbb{1}, \sigma^0 \rangle := 1$ by definition.
 - ▶ $\langle \mathbb{1}, \sigma^k \rangle := \int_{\sigma^k} 1$ for $k \geq 1$ (length / area / volume / ...)

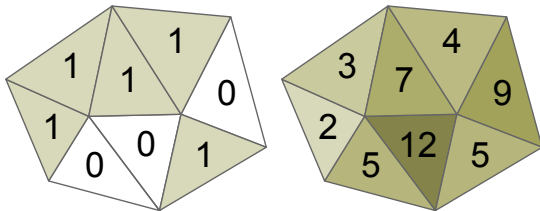
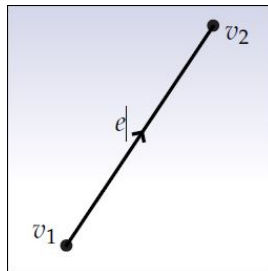


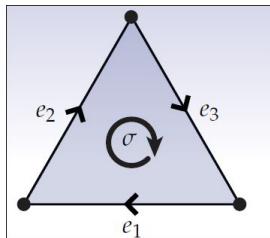
Figure: The scalar product $\langle \omega^2, c^2 \rangle$ integrates the co-chain ω^2 over the 2-chain c^2 as $(3, 7, 4, 9, 2, 5)^T (1, 1, 1, 0, 0, 1) = 21$

- ▶ ϕ - primal 0-form (vertices)
- ▶ $d\phi$ - primal 1-form (edges)



$$(\hat{d}\phi)_e = \int_e d\phi = \int_{\partial e} \phi = \hat{\phi}(v_2) - \hat{\phi}(v_1)$$

- ▶ α - primal 1-form (edges)
- ▶ $d\alpha$ - primal 2-form (triangles)

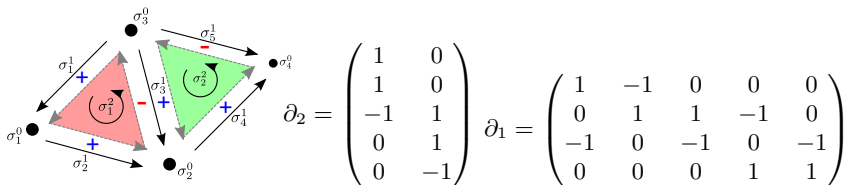


$$(\hat{d}\alpha)_\sigma = \int_\sigma d\alpha = \int_{\partial\sigma} \alpha = \sum_{i=1}^3 \int_{e_i} \alpha = \sum_{i=1}^3 \hat{\alpha}_i$$

In general: discrete exterior derivative is coboundary operator for cochains.



- ▶ For an oriented simplicial complex, we can define (signed) boundary operators ∂_k , which map k -simplices to the $(k - 1)$ -simplices on their boundary
- ▶ The boundary operator $\partial_k : C^k \rightarrow C^{k-1}$ maps k -chains to $k - 1$ chains, which are the combined boundary of the simplices in the chain. Inner edges cancel out.
- ▶ The discrete boundary operator is the *signed* adjacency matrix $\partial_k \in \mathbb{R}^{N_{k-1} \times N_k}$.
 - ▶ ∂_2 maps triangles to their boundary edges
 - ▶ ∂_1 maps edges to their endpoints
 - ▶ The signs correspond to the relative orientations between the simplex and the boundary simplices
- ▶ 2D example:



- ▶ Note the difference to signed adjacency matrices: The relative orientation of edges is not related to neighbor triangles, so boundary edges may have a negative sign.



Boundary Operator



- ▶ Inner $(k - 1)$ -simplices cancel out each other as they are summed two times with opposite relative orientation.
- ▶ We have $\partial_{k-1}\partial_k = 0$ as the boundary of a k chain is always closed and thus does not have a boundary itself.
- ▶ Boundary edges can have a negative orientation.

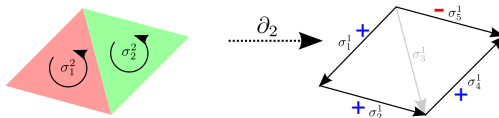


Figure: The boundary operator applied to adjacent triangles maps to a 1-chain containing the outer boundary of the two triangles.

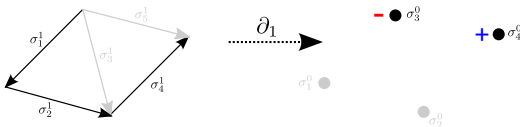


Figure: The boundary operator applied to a 1-chain of three connected edges maps to a 0-chain containing the two endpoints of the path.



Defining a discrete differential operator



A discrete differential operator can be defined using Stokes' theorem“

$$\int_{\sigma} d\hat{\omega} = \int_{\partial\sigma} \hat{\omega}$$

The domain is given by a k -chain vector c^k and the integral over $d\omega^{k+1}$ can be evaluated using the boundary operator ∂

$$\langle d\omega^k, c^{k+1} \rangle = \langle \omega^k, \partial c^{k+1} \rangle = \sum_{\sigma^k \in \partial c_{k+1}} \langle \omega^k, \sigma^k \rangle$$

As the discrete operators are matrices and discrete co-chains are vectors we have

$$\langle \omega^k, \partial c^{k+1} \rangle = \langle \partial^T \omega^k, c^{k+1} \rangle$$

so the discrete differential operator is the transpose of the discrete boundary operator:

Discrete Differential Operator

$$\mathbf{d}_k := \partial_{k+1}^T$$

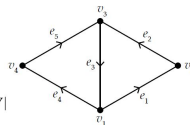
- ▶ To build the exterior derivative on 0-forms, we first need to assign an index to each vertex and each edge
 - ▶ A discrete 0-form is a list of $|V|$ values (one per vertex)
 - ▶ A discrete 1-form is a list of $|E|$ values (one per edge)
- ▶ The discrete exterior derivative d_0 is therefore a $|E| \times |V|$ matrix, taking values at vertices to values at edges

Example.

$$\phi \in \mathbb{R}^{|V|}$$

$$\alpha \in \mathbb{R}^{|E|}$$

$$d_0 \in \mathbb{R}^{|E| \times |V|}$$



$$\begin{array}{c} e_1 \\ e_2 \\ e_3 \\ e_4 \\ e_5 \end{array} \begin{bmatrix} v_1 & v_2 & v_3 & v_4 \\ -1 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ 1 & 0 & -1 & 0 \\ -1 & 0 & 0 & 1 \\ 0 & 0 & 1 & -1 \end{bmatrix} \begin{bmatrix} \phi_1 \\ \phi_2 \\ \phi_3 \\ \phi_4 \end{bmatrix} = \begin{bmatrix} \alpha_1 \\ \alpha_2 \\ \alpha_3 \\ \alpha_4 \\ \alpha_5 \end{bmatrix}$$

$d_0 \qquad \qquad \phi \qquad \qquad \alpha$

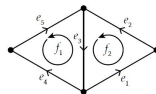
- To build the exterior derivative on 1-forms, we first need to assign an index to each edge and each face
 - A discrete 1-form is a list of $|E|$ values (one per edge)
 - A discrete 2-form is a list of $|F|$ values (one per face (triangle))
- The discrete exterior derivative d_1 is therefore a $|F| \times |E|$ matrix, taking values at vertices to values at edges

Example.

$$\alpha \in \mathbb{R}^{|E|}$$

$$\omega \in \mathbb{R}^{|F|}$$

$$d_1 \in \mathbb{R}^{|F| \times |E|}$$



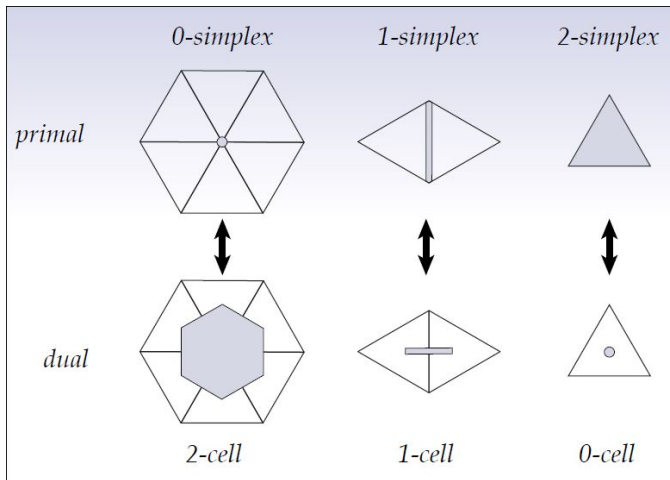
$$d_1 = \begin{matrix} f_1 & \begin{bmatrix} e_1 & e_2 & e_3 & e_4 & e_5 \\ 0 & 0 & -1 & -1 & -1 \\ 1 & 1 & 1 & 0 & 0 \end{bmatrix} \\ f_2 & \end{matrix} \begin{bmatrix} \alpha_1 \\ \alpha_2 \\ \alpha_3 \\ \alpha_4 \\ \alpha_5 \end{bmatrix} = \begin{bmatrix} \omega_1 \\ \omega_2 \end{bmatrix} \omega$$

- ▶ By definition, the discrete exterior derivative satisfies a very important property:
- ▶ Taking the smooth exterior derivative and then discretizing yields the same result as discretizing and then applying the discrete exterior derivative.
- ▶ d - smooth exterior derivative
- ▶ $\hat{d}$ - discrete exterior derivative
- ▶ $\int$ - de Rham map (discretization)
- ▶ α - smooth k -form
- ▶ $\hat{\alpha}$ - discrete k -form
- ▶ $d\alpha$ - smooth $k + 1$ -form
- ▶ $\hat{d}\hat{\alpha}$ - discrete $k + 1$ -form

$$\begin{array}{ccc}
 \alpha & \xrightarrow{d} & d\alpha \\
 \int \downarrow & & \downarrow \int \\
 \hat{\alpha} & \xrightarrow{\hat{d}} & \hat{d}\hat{\alpha}
 \end{array}$$

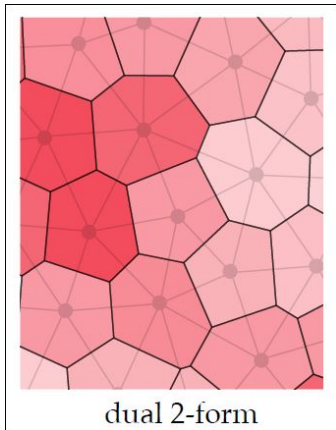


Reminder: Poincaré Duality



Consider the dual K^* of a manifold simplicial complex K .

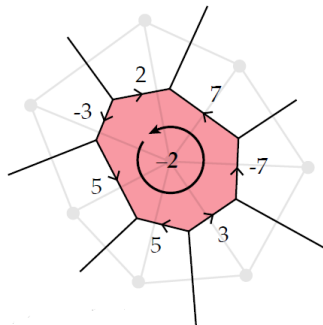
- ▶ Just as a discrete differential k -form was a value per k -simplex, a dual discrete differential k -form is a value per k -cell:
 - ▶ a dual 0-form is a value dual vertex
 - ▶ a dual 1-form is a value per dual edge
 - ▶ a dual 2-form is a value per dual cell
- ▶ Taking the smooth exterior derivative and then discretizing yields the same result as discretizing and then applying the discrete exterior derivative.





Discrete exterior derivative on dual k -forms works in essentially the same way as for primal forms:

- ▶ To get the derivative on a $(k+1)$ -cell, sum up values on each k -cell along its boundary
- ▶ Sign of each term in the sum is determined by relative orientation of $(k+1)$ -cell and k -cell

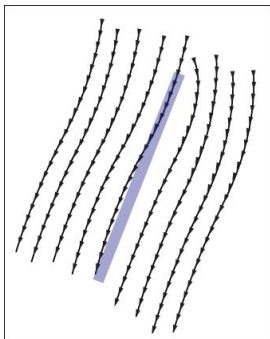


(a)

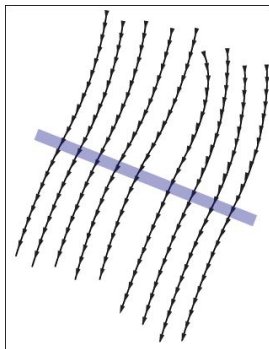
$$-7 + 7 - 2 + (-3) + 5 - 5 + 3 = -2$$

Example:

- ▶ Let α be a dual discrete 1-form (one value per dual edge)
- ▶ Then $d\alpha$ is a value per 2-cell, obtained by summing over dual edges



(a) Primal 1-form (circulation)



(a) Dual 1-form (flux)

Example:

- ▶ To capture Hodge duality in the discrete setting, we'll need to define a dual mesh.
- ▶ In general, the dual of an n -dimensional simplicial mesh identifies every k -simplex in the primal (i.e., original) mesh with a unique $(n - k)$ -cell in the dual mesh.
- ▶ Note, however, that the dual cells are not always simplices!



The Hodge star operator



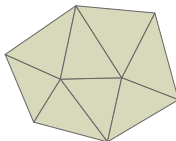
We would like to map discrete differential forms ω^k which are defined on primal simplices to dual discrete differential forms $\omega_d^{n-k} = \star \omega^k$ defined on the corresponding dual cells $\star \sigma^k$, such that the average values over the areas agree:

$$\frac{1}{|\star \sigma^k|} \langle \star \omega^k, \star \sigma^k \rangle = \frac{1}{|\sigma^k|} \langle \omega^k, \sigma^k \rangle$$

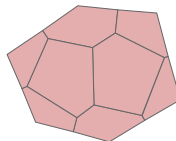
where $|\cdot|$ denotes the volume of the k -simplex or k -cell.

It especially holds that the sum of the differential form evaluated on all simplices is the same as the sum of the dual form evaluated on all dual cells:

$$\sum_{\sigma_{\text{dual}}^k \in K_d^k} \langle \star \omega^k, \sigma_{\text{dual}}^{n-k} \rangle = \sum_{\sigma^k \in K^k} \langle \omega^k, \sigma^k \rangle$$



(a) The primal mesh



(b) The dual mesh



The Hodge star operator



To calculate the integration of the differential form $\star \hat{\omega}^k$ over the dual cell $\star \sigma_i^k$, we can rearrange the equation:

$$\underbrace{\langle \star \omega^k, \star \sigma_i^k \rangle}_{\text{dual}} = \underbrace{\frac{|\star \sigma_i^k|}{|\sigma_i^k|}}_{\text{Hodge Star}} \underbrace{\langle \omega^k, \sigma_i^k \rangle}_{\text{primal}}$$

The Hodge star operator for discrete k -forms can then be defined as diagonal matrix

$$\star_k := \begin{pmatrix} \frac{|\star \sigma_1^k|}{|\sigma_1^k|} & & \\ & \ddots & \\ & & \frac{|\star \sigma_{N_k}^k|}{|\sigma_{N_k}^k|} \end{pmatrix}$$

so we can calculate the dual discrete differential form $\omega_{\text{dual}}^{n-k} = \star_k \omega^k$ as a matrix-vector product.

For $k = 0$ and $k = n$ we have an additional sign rules $s := (-1)^{n-1} \text{sgn}(\partial(\star \sigma^0), \star \sigma^1)$ and $s := (-1)^{n-1} \text{sgn}(\star \sigma^n, \partial(\star \sigma^{n-1}))$ to account for the orientation.



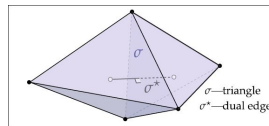
Definition Diagonal Hodge Star

Let Ω_k and Ω_{n-k}^* denote the primal k -forms and dual $(n - k)$ -forms respectively on a n -dimensional simplicial manifold M . The diagonal Hodge star is a map $\star : \Omega_k \rightarrow \Omega_{n-k}^*$ determined by

$$\star \alpha(\sigma^\star) = \frac{|\sigma^\star|}{|\sigma|} \alpha(\sigma)$$

for each k -simplex $\sigma \in M$, where $\sigma^\star$ is the corresponding dual cell, and $|\cdot|$ denotes the volume of a simplex or cell.

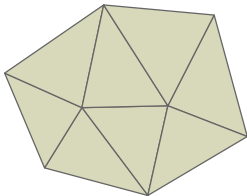
Key idea: divide by primal area, multiply by dual area.



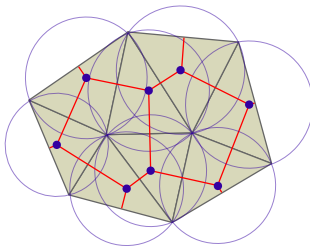
(a) Dual 1-form (flux)



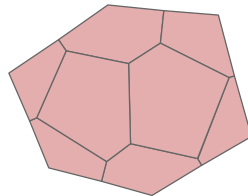
- ▶ DEC uses the *circumcenter*, i.e., the center of the circle through the vertices of a triangle.
- ▶ For general simplicial complexes: The dual vertex of a k -simplex in a k -dimensional simplicial complex is the center of the n -hypersphere through its points
- ▶ The intersections of the dual edges of boundary edges with the corresponding primal edges are added as additional dual vertices.
- ▶ The edges between a dual vertex on the boundary and the two adjacent primal vertices are added to close the dual cell at the boundary.



(a) The primal mesh



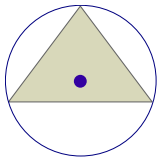
(b) To construct a dual mesh, the circumcenters of adjacent triangles are connected



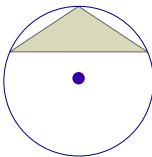
(c) The dual mesh is a cell complex



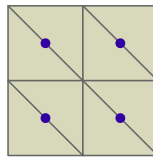
- ▶ Advantages of a circumcentric dual mesh:
 - ▶ The primal and dual edges are orthogonal.
 - ▶ The construction allows for recursive calculation of the lengths/areas/volumes.
 - ▶ The related DEC operators are *sparse* matrices.
- ▶ Disadvantages of a circumcentric dual mesh:
 - ▶ The triangles have to be *well-centered* (Hirani 2003).
 - ▶ A triangle is well-centered, when it contains its own circumcenter, and a mesh is well-centered when all its triangles are well-centered.
 - ▶ Most meshes are *not* well-centered.
- ▶ Problems with triangles which are not well-centered
 - ▶ The dual vertex lies outside.
 - ▶ The dual edge lies outside the triangle or is flipped.
 - ▶ Two dual vertices can lie at the same point, causing a *degenerate* edge of length 0.
 - ▶ Problem: Some operators need to divide by the dual edge length.
 - ▶ Meshes with degenerate dual edges are not rare. A triangulated grid is a prime example.



(a) A well-centered triangle



(b) Not a well-centered triangle



(c) A mesh with degenerate dual edges

We have defined the differential operator $\mathbf{d}_k : \Omega^k \rightarrow \Omega^{k+1}$, which maps k -forms to $(k+1)$ -forms. Now we want to define a codifferential operator $\delta_{k+1} : \Omega^{k+1} \rightarrow \Omega^k$, which maps $(k+1)$ -forms back to k -forms.

We define the operator δ_k using the Hodge star and the dual differential operators as

$$\delta_k \omega^k = (\star_{k-1}^{-1} \mathbf{d}_{n-k}^{\text{dual}} \star_k) \omega^k$$

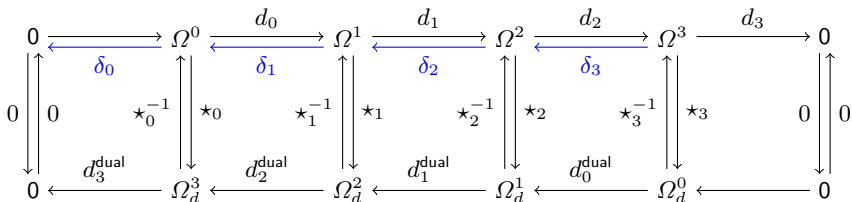


Figure: The DeRham complex shows the relation between primal and dual differential operators for a 3D simplicial complex



Using the differential and codifferential operators, we can now define a discrete Laplace operator by

$$L_k := \mathbf{d}_{k-1}\delta_k + \delta_{k+1}\mathbf{d}_k$$

In particular we have

$$\begin{aligned} L_0 &= \delta_1 \mathbf{d}_0 &= \star_0^{-1} \mathbf{d}_1^{\text{dual}} \star_1 \mathbf{d}_0 \quad \text{and} \\ L_n &= \mathbf{d}_{n-1} \delta_n &= \mathbf{d}_{n-1} \star_{n-1}^{-1} \mathbf{d}_0^{\text{dual}} \star_n \end{aligned}$$

because $\mathbf{d}_n = 0$ and $\delta_0 = 0$.

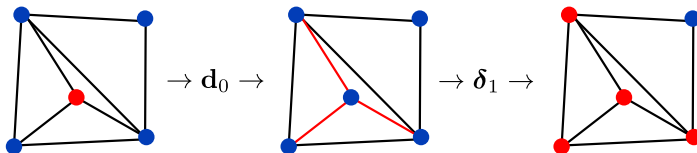


Figure: The differential operator $\mathbf{d}_0$ uses the values at the red vertices to calculate the directed derivative on the adjacent edges. Afterward, the codifferential operator δ_1 maps the values on the edges back to values on adjacent vertices.



- ▶ The operator $\flat$ maps vector fields to 1-forms and the operator $\sharp$ maps 1-forms to vector fields. For the details of the different kinds of $\flat$ and $\sharp$ operators, see A. Hirani, 2003.
- ▶ $\mathbf{d}\omega^0$ is the *directional* derivative along the edges. This allows for interpolation of a tangential gradient field $\nabla\omega^0 = (\mathbf{d}\omega^0)^\sharp$.
- ▶ The divergence of 1-form $X^\flat$ of a tangential vector field is $\nabla \cdot X = \star \mathbf{d} \star X^\flat = \delta X^\flat$.
- ▶ This matches our definition of the Laplace operator: $\Delta\omega^0 = \nabla \cdot \nabla\omega^0 = \delta \mathbf{d}\omega^0$
- ▶ The curl is defined as a vector field on the dual edges: $\nabla \times X = (\star \mathbf{d} X^\flat)^\sharp$

Common vector identities hold:

$$\begin{aligned}\text{div} \circ \text{curl } X &= \nabla \cdot \nabla \times X = \star \mathbf{d} \star (\star \mathbf{d} X^\flat)^\sharp = 0 \\ \text{div} \circ \text{grad } \omega^0 &= \nabla \times \nabla \omega^0 = (\star \mathbf{d} (\mathbf{d}\omega^0)^\flat)^\sharp = 0\end{aligned}$$

because $\mathbf{d}\mathbf{d} = 0$ and $\mathbf{d} \star \star \mathbf{d} = 0$.



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 - ▶ **Current state of the art.**
- ▶ Using concyclic polygons to allow for dual meshes with edge lengths, which are exactly 0. (Schier and Klein, 2023)



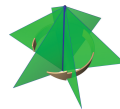
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Conclusion: DEC with circumcentric duality is possible for all meshes when using the method of Mohamed et al.



- ▶ Solving PDEs on discrete domains (given meshes) instead of discretized domains.
- ▶ In computer graphics, we often start with a given mesh anyway.
 - ▶ Often someone creates a mesh, e.g., for movies and there is no real object.
- ▶ The values of interest are located at mesh elements.
- ▶ The points of interest often are related to the quantity that is associated with the element:



(a) Flux is defined at the faces (b) Divergence is defined in the cells (c) Curl is defined at the edges

Figure: Physical quantities in a fluid simulation and their location at the mesh elements in a tetrahedral mesh.

- ▶ Relation to finite differences and finite volumes schemes: Equivalence to FDM and FVM can be proven².

²A. Schier, C. Rieger, et al. "Upwind Schemes for Scalar Advection-Dominated Problems in the Discrete Exterior Calculus". In: *Transport Processes at Fluidic Interfaces*. Ed. by D. Bothe and A. Reusken. Springer International Publishing, 2017, pp. 145–175.



- ▶ DEC can be used to derive the FDM/FVM schemes on triangle meshes.
- ▶ DEC is *not* directly related to FEM (Finite Element methodes).
 - ▶ For FEM there are approaches like finite element exterior calculus (FEEC). (Arnold et al., 2006)
- ▶ Operators are defined uniquely, i.e, there are no parameters that need to be chosen for the scheme³.
- ▶ The scheme works the same for triangle meshes, tetrahedral meshes, and generalizes to simplicial complexes in higher dimensions.
- ▶ Elegant formulations for common discrete operators like ∇ , Δ , divergence, and curl.
- ▶ Separation of *exact* operators and *approximate* operators.
- ▶ Discrete Exterior Calculus provides discrete k -forms, which are paired with k -simplices (instead of k -vectors).
 - ▶ For DEC it is easier to think of k -forms as functions defined for simplices.
- ▶ DEC can be used to explain some well-known discrete operators.
 - ▶ Example: The standard DEC Laplacian is equivalent to the cotangent Laplacian⁴.

³In contrast to the vast amount of different FEM discretizations without an easy criterion which one is suitable for the problem at hand.

⁴U. Pinkall and K. Polthier. "Computing discrete minimal surfaces and their conjugates". In: *Experimental mathematics* 2.1 (1993), pp. 15–36.



Solving Partial Differential Equations (PDEs) on meshes

- ▶ Some common methods
 - ▶ Finite Differences (FDM) approximates values on vertices of a grid or mesh.
 - ▶ Finite Volumes (FVM) approximates integrated values inside faces/cells.
 - ▶ Finite Elements (FEM) uses the weak formulation of the problem.



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 - ▶ Finite Elements (FEM) uses the weak formulation of the problem.
- ▶ These methods *discretize* a given domain, e.g., using a grid or by triangulating them.
 - ▶ Different discretizations for different applications.
- ▶ The methods provide an *approximation* of the real solution.
 - ▶ The accuracy depends on the mesh resolution and mesh quality.
 - ▶ Involved quantities are approximated at the points of interest.
 - ▶ The methods work best with meshes optimized for the particular method.

Discrete Exterior Calculus (DEC)

- ▶ Described by Anil N. Hirani in 2003⁵.
- ▶ Uses *existing* meshes, so it has no discretization step.
- ▶ Values are associated with mesh elements (vertices, edges, faces).
- ▶ Calculations are *exact* at the mesh elements.
- ▶ Integral theorems like Stokes' theorem, Green's theorem and the divergence theorem hold *by definition*.
- ▶ But there is also no interpolation in pure DEC, to interpolate values, e.g., on points inside triangles.
 - ▶ There are some interpolation schemes like Whitney forms.
 - ▶ But they again introduce an approximation error.
- ▶ The strength of DEC is its exactness on discrete elements.
- ▶ When two operations cancel out each other, DEC makes sure that they actually cancel out without a numeric error, i.e., $(O^{-1} \circ O) = \text{Id}$ holds *exactly*.

⁵A. Hirani. "Discrete exterior calculus". PhD thesis. California Institute of Technology, 2003.



Example Applications

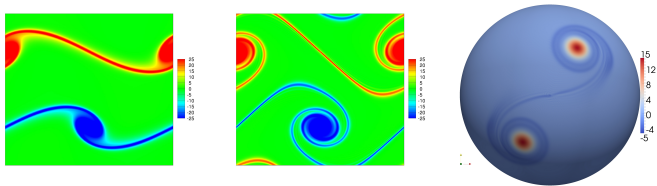


Figure: M. S. Mohamed et al. “Discrete exterior calculus discretization of incompressible Navier–Stokes equations over surface simplicial meshes”. In: *Journal of Computational Physics* 312 (2016), pp. 175–191

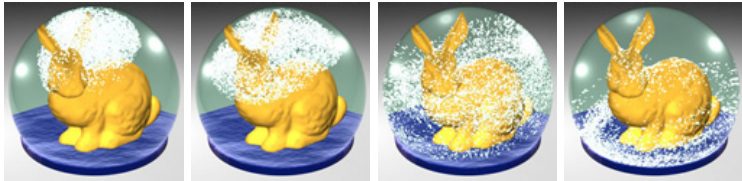


Figure: S. Elcott et al. “Stable, circulation-preserving, simplicial fluids”. In: *ACM Transactions on Graphics (TOG)* 26.1 (2007), 4–es



Example Applications

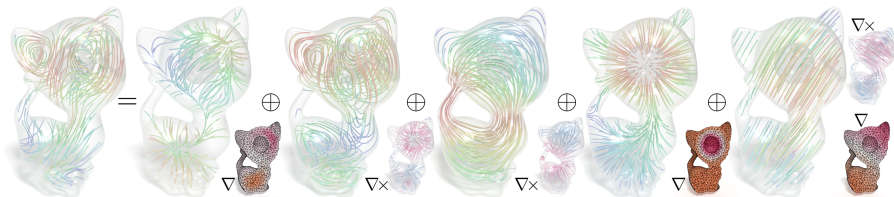


Figure: R. Zhao et al. “3D hodge decompositions of edge- and face-based vector fields”. In: *ACM Transactions on Graphics (TOG)* 38.6 (2019), pp. 1–13



Figure: K. Crane et al. “Geodesics in heat: A new approach to computing distance based on heat flow”. In: *ACM Transactions on Graphics (TOG)* 32.5 (2013), pp. 1–11

N. Sharp et al. “The vector heat method”. In: *ACM Transactions on Graphics (TOG)* 38.3 (2019), pp. 1–19



There are more DEC operators, which are not covered by this lecture:

- ▶ Wedge Product $\wedge$: Maps a k -form and ℓ -form to a $(k + \ell)$ -form
- ▶ Interior Product
- ▶ Inner Product
- ▶ Interpolation of forms using Whitney forms
- ▶ Flat Operator $\flat$: Maps vector fields to 1-forms.
- ▶ Sharp Operator $\sharp$: Maps 1-forms to vector fields, e.g., $\nabla \hat{\omega} = (\mathbf{d}\omega)^\sharp$.

Real world example⁶: The incompressible Navier-Stokes equations

$$\frac{\partial u}{\partial t} - \nu \nabla^2 u + u \cdot \nabla u + \nabla p = 0$$

$$\nabla \cdot u = 0$$

Can be discretized with the DEC scheme (Mohamed et al., 2018) as

$$\frac{\partial \mathbf{u}^b}{\partial t} - \nu \star \mathbf{d} \star \mathbf{d}\mathbf{u}^b + \star(\mathbf{u}^b \wedge \star \mathbf{d}\mathbf{u}^b) + \mathbf{d}p^d = 0 \quad [\text{with } p^d = p + \frac{1}{2}(\mathbf{u}^b(\mathbf{u}))]$$

$$\star \mathbf{d} \star \mathbf{u}^b = 0$$

⁶This will not be in the exam



- ▶ A. Hirani. “Discrete exterior calculus”. PhD thesis. California Institute of Technology, 2003
- ▶ M. Desbrun et al. “Discrete exterior calculus”. In: *arXiv preprint math/0508341* (2005)
- ▶ Finite Element Exterior Calculus (FEEC). (Arnold et al., 2006)
- ▶ K. Crane. “Discrete differential geometry: An applied introduction”. In: *Notices of the AMS, Communication* (2018), pp. 1153–1159
- ▶ Discrete Differential Geometry: An Applied Introduction (Siggraph Asia 2008 Course Notes)⁷
 - ▶ Beware: The 2006 course notes contain some errors.
- ▶ Subdivision Exterior Calculus (SEC). (Goes et al., 2016)
- ▶ Discrete Differential Geometry lecture by Keenan Crane
<http://brickisland.net/DDGSpring2020/>

⁷<http://ddg.math.uni-goettingen.de/pub/SiggraphAsia2008DDGCourse.pdf>



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